CSE 717 — Information Security

Topic 8: RSA — Key Generation & Encryption/Decryption Numericals

Complete Past Paper Solutions: 2020–2024

Source: Lc#6 pp16–21 (RSA Algorithm Steps + worked examples)

HIGHEST-YIELD TOPIC: 5/5 years, 5–16 marks. Every Topic-8 question 2020–2024 included. Zero omissions.

Contents

Quick Reference — The RSA Pipeline (CHEAT SHEET)

CONFIRMED EVERY YEAR — builds directly on Topic 7 (Extended Euclidean + I

Every past-paper RSA question is the SAME 8-step pipeline with different small numbers. "Given e, n find d" (2020 & 2021, $e = 31, n = 3599$) is a near-certain repeat style. Always use **square-and-multiply** for $M^e \bmod n$ — never multiply out the full power.

The 8-Step RSA Algorithm (Lc#6 p16)

1. Select two primes $p \neq q$.
2. Compute $n = p \times q$.
3. Compute $\phi(n) = (p-1)(q-1)$.
4. Select e such that $\gcd(e, \phi(n)) = 1$ and $1 < e < \phi(n)$.
5. Compute $d = e^{-1} \bmod \phi(n)$, i.e. $ed \equiv 1 \pmod{\phi(n)}$ (Extended Euclidean — Topic 7 Cheat Sheet B).
6. **Public key** $= \{e, n\}$ (shared with everyone). **Private key** $= \{d, n\}$ (kept secret).
7. **Encrypt:** $C = M^e \bmod n$, where $M < n$.
8. **Decrypt:** $M = C^d \bmod n$.

Worked example from slides (Lc#6 pp17–19): $p = 13, q = 11$

$$n = 143, \quad \phi(n) = 12 \times 10 = 120, \quad e = 13 \ (\gcd(13, 120) = 1)$$

$$13d \equiv 1 \pmod{120} \Rightarrow d = 37 \quad (\text{check: } 13 \times 37 = 481 = 4 \times 120 + 1)$$

$$\text{Public key} = \{13, 143\}, \quad \text{Private key} = \{37, 143\}$$

$$\text{Encrypt } P = 13 : \quad C = 13^{13} \bmod 143 = 52$$

$$\text{Decrypt } C = 52 : \quad P = 52^{37} \bmod 143 = 13\checkmark$$

Square-and-Multiply — the ONLY way to compute $M^e \bmod n$ by hand

Write e in binary. Scan bits left to right, starting with result = 1:

- Always: **square** the result mod n.
- If the current bit is 1: also **multiply** by M mod n.

Example: $e = 17 = 10001_2$ needs only 4 squarings + 1 multiply (instead of 16 multiplications).

2024 — Q2(c): Encrypt "DATA" (3); Q2(d): Decrypt $c = 17$ (3)

Question — 2024 Q2(c) & Q2(d)

(c) Encrypt the message "DATA" using the RSA cryptosystem with $p = 61$, $q = 53$, and $e = 17$. [3]

(d) An RSA-encrypted message consists of the ciphertext block $c = 17$. The RSA system parameters are $p = 61$, $q = 53$, $e = 17$. Decrypt the ciphertext and find the plaintext letter. [3]

Part (c): Encrypt "DATA"

Answer — 2024 Q2(c)

Step 1–3: $n = p \times q = 61 \times 53 = \boxed{3233}$, $\quad \phi(n) = (60)(52) = \boxed{3120}$.
Step 4: $e = 17$, check $\gcd(17, 3120) = 1$ ✓
Encoding: use each letter's ASCII code as the plaintext block M (each $M < 3233$, so each letter is its own block). $D = 68$, $A = 65$, $T = 84$, $A = 65$.
Compute $C = M^{17} \bmod 3233$ via square-and-multiply ($17 = 10001_2 \to 4$ squarings, multiply by M on bits 1 and 5):

Letter (M)	after bit 1	after bit 2 (sq)	after bit 3 (sq)	after bit 4 (sq)	after bit 5 (sq+$\times M$)
D (68)	68	1391	1547	789	$\boxed{1759}$
A (65)	65	992	1232	1547	$\boxed{2790}$
T (84)	84	590	2169	546	$\boxed{2159}$
A (65)	65	992	1232	1547	$\boxed{2790}$

Ciphertext blocks: "DATA" $\rightarrow$ $\boxed{1759,\ 2790,\ 2159,\ 2790}$.
(This is the standard Stallings textbook example: $A = 65 \rightarrow C = 2790$.)

Part (d): Decrypt $c = 17$

Answer — 2024 Q2(d)

Find d: need $d = e^{-1} \bmod \phi(n) = 17^{-1} \bmod 3120$ (Extended Euclidean, Topic 7 Cheat Sheet B).

$$17 \times 2753 = 46801 = 15 \times 3120 + 1 \;\Rightarrow\; d = \boxed{2753}$$

Decrypt: $M = C^d \bmod n = 17^{2753} \bmod 3233$. Using repeated square-and-multiply on the 12-bit exponent $2753 = 101011000001_2$:

$$M = 17^{2753} \bmod 3233 = \boxed{3170}$$

Note on "plaintext letter": 3170 falls outside the printable ASCII range (0–255), so this particular block does not map back to a single readable character under the ASCII encoding used in part (c). **The mark is for the correct method** (compute d via Extended Euclidean, then $C^d \bmod n$ via square-and-multiply) — show all steps even if the final number isn't a "nice" letter. If the examiner expects a letter, state the numeric result 3170 and note it lies outside the A–Z/ASCII range.

2023 — Section B Q7(c): Encrypt "CU" (3); Q7(d): Decrypt (2)

Question — 2023 Section B Q7(c) & Q7(d)

(c) Encrypt the message "CU" using the RSA cryptosystem with $p = 43$ and $q = 59$. [3]

(d) Consider an encrypted message "0981 0461". What is the decrypted message if it was encrypted using the RSA cryptosystem with $p = 43$ and $q = 59$? [2]

Note: e is not legible in the scanned paper for this question

The value of e is missing from the source PDF for Q7(c)/(d) (likely a scan/crop issue). n and $\phi(n)$ depend only on p, q and are fixed; e must still satisfy $\gcd(e, \phi(n)) = 1$. Below, the **full pipeline** is demonstrated with a self-chosen valid $e = 13$ so the worked example is internally consistent and exam-ready — the *method* is identical regardless of which valid e the original paper used.

Part (c): Encrypt "CU"

Answer — 2023 Section B Q7(c)

Steps 1–3: $n = p \times q = 43 \times 59 = \boxed{2537}$, $\phi(n) = (42)(58) = \boxed{2436}$.
Step 4: choose $e = 13$. Check $\gcd(13, 2436) = 1$ ✓ (a valid public exponent).
Encoding: ASCII codes $C = 67$, $U = 85$ (each < 2537, one block per letter).
Compute $C = M^{13} \bmod 2537$ **via square-and-multiply** ($13 = 1101_2 \to$ bits: 1,1,0,1):

Letter (M)	after bit 1	after bit 2 (sq+$\times M$)	after bit 3 (sq)	after bit 4 (sq+$\times M$)
C (67)	67	1397	656	$\boxed{2044}$
U (85)	85	171	1334	$\boxed{1246}$

Ciphertext: "CU" $\to \boxed{2044,\ 1246}$ (with $e = 13,\ n = 2537$)

Part (d): Decrypt the ciphertext

Answer — 2023 Section B Q7(d)

Find d**:** $d = e^{-1} \bmod \phi(n) = 13^{-1} \bmod 2436$ (Extended Euclidean):

$$13 \times 937 = 12181 = 5 \times 2436 + 1 \;\Rightarrow\; d = \boxed{937}$$

Decrypt $M = C^d \bmod n = C^{937} \bmod 2537$ for each block:

$$2044^{937} \bmod 2537 = \boxed{67} = \text{‘C’}, \qquad 1246^{937} \bmod 2537 = \boxed{85} = \text{‘U’}$$

$\Rightarrow$ plaintext recovered = "CU" ✓
The paper's actual ciphertext "0981 0461" was produced with the original (unknown) e – the same Extended-Euclidean-then-square-and-multiply procedure applies verbatim once e (and hence d) is known. Demonstrating the full round-trip (encrypt then decrypt back to the same letters) earns full method marks.

2022 — Q6: Encrypt $M = 3$, choose $e < 10$ (3+3+2+8=16)

Question — 2022 Q6

Encrypt the message $M = 3$ via RSA with $p = 13$, $q = 7$. Choose an $e < 10$ and show all steps. Explain why an eavesdropper (Charlie), who intercepts the ciphertext, cannot decrypt the message without the private key. [3+3+2+8=16]

Answer — 2022 Q6

Steps 1–3: $n = p \times q = 13 \times 7 = \boxed{91}$, $\phi(n) = (12)(6) = \boxed{72}$.
Step 4 — choose $e < 10$: need $\gcd(e, 72) = 1$ with $1 < e < 10$. Candidates: $e = 5$ ($\gcd(5, 72) = 1$ ✓) or $e = 7$ ($\gcd(7, 72) = 1$ ✓). Take $\boxed{e = 5}$.
Step 5 — find d: $d = 5^{-1} \bmod 72$.

$$5 \times 29 = 145 = 2 \times 72 + 1 \;\Rightarrow\; d = \boxed{29}$$

Step 6: Public key $= \{5, 91\}$, Private key $= \{29, 91\}$.
Step 7 — encrypt $M = 3$: $C = 3^5 \bmod 91$. $5 = 101_2$:

$$3 \xrightarrow{\text{sq}} 9 \xrightarrow{\text{sq}\times 3} 61 \quad \Rightarrow \quad C = \boxed{61}$$

Step 8 — decrypt (verification): $M = 61^{29} \bmod 91 = \boxed{3}$ ✓

Why Charlie can't decrypt without the private key:

- Charlie sees the public key $\{e, n\} = \{5, 91\}$ and the ciphertext $C = 61$, but **not** $d = 29$.
- Decryption requires $M = C^d \bmod n$ — without d, Charlie must either:
  - **Factor** $n = 91$ into $p = 13, q = 7$ to recompute $\phi(n)$ and then d — for small textbook numbers this is feasible, but for real RSA (n with 300+ digit primes) factoring is **computationally infeasible** (this is the RSA hardness assumption, based on the integer factorization problem).
  - **Brute-force every possible** $M < n$, computing $M^e \bmod n$ and comparing to C — exponential search space for real key sizes.
- Even knowing e and n, computing $d = e^{-1} \bmod \phi(n)$ requires knowing $\phi(n)$, which requires knowing p and q — and $n = pq$ does not reveal p, q without factoring.
- **Conclusion:** the security of RSA rests entirely on the fact that factoring a large n into its two prime factors is practically impossible with current computing power, so d remains secret even though e and n are public.

2021 & 2020 — $p = 3, q = 11, e = 7, M = 2$ (Encrypt+Decrypt); $e = 31, n = 3599$ find d

Question — 2021 Section B Q2(a)&(b), and identical 2020 Q6(a)&(b)

(a) Encrypt and decrypt using RSA with $p = 3, q = 11, e = 7, M = 2$. [2020: 4.75] [2021: 2.75]

(b) Given $e = 31,\ n = 3599$, find the private key d. [2+2 (both years)]

Part (a): RSA with $p = 3, q = 11, e = 7, M = 2$

Answer — 2021 B2(a) / 2020 Q6(a)

Steps 1–3: $n = 3 \times 11 = \boxed{33}$, $\phi(n) = (2)(10) = \boxed{20}$.
Step 4: $e = 7$, check $\gcd(7, 20) = 1$ ✓
Step 5 — find d: $d = 7^{-1} \bmod 20$.

$$7 \times 3 = 21 = 1 \times 20 + 1 \Rightarrow d = \boxed{3}$$

Step 6: Public key $= \{7, 33\}$, Private key $= \{3, 33\}$.
Step 7 — encrypt $M = 2$: $C = 2^7 \bmod 33$. $7 = 111_2$:

$$2 \xrightarrow{\text{sq}\times 2} 8 \xrightarrow{\text{sq}\times 2} 29 \quad \Rightarrow \quad C = \boxed{29}$$

Step 8 — decrypt: $M = 29^3 \bmod 33 = \boxed{2}$ ✓

Part (b): Given $e = 31$, $n = 3599$, find d

Answer — 2021 B2(b) / 2020 Q6(b)

Factor $n = 3599$ into its two primes (try small primes upward):

$$3599 = 59 \times 61 \quad \Rightarrow \quad p = 59,\ q = 61$$

Compute $\phi(n)$**:**

$$\phi(3599) = (59 - 1)(61 - 1) = 58 \times 60 = \boxed{3480}$$

Find $d = 31^{-1} \bmod 3480$ via Extended Euclidean:

$$31 \times 3031 = 93961 = 27 \times 3480 + 1 \Rightarrow d = \boxed{3031}$$

Private key $= \{3031,\ 3599\}$. *(This exact pair $e = 31, n = 3599$ repeats verbatim in 2020 & 2021 — memorize $n = 59 \times 61$, $\phi(n) = 3480$, $d = 3031$.)*

Exam Strategy Summary

High-Yield Checklist for RSA (5/5 years, 5–16 marks — LARGEST single-topic bloc

1. **Always small primes** in the range $\{3, 7, 11, 13, 17, 31, 43, 53, 59, 61\}$ — never need a calculator for $n = p \times q$ or $\phi(n) = (p - 1)(q - 1)$.
2. **The 8-step pipeline never changes:** $n \to \phi(n) \to$ pick/verify $e \to d = e^{-1} \bmod \phi(n)$ (Extended Euclidean, Topic 7) $\to C = M^e \bmod n \to M = C^d \bmod n$ (square-and-multiply).
3. **"Given e, n find d"** is a near-certain repeat: memorize $e = 31, n = 3599 \Rightarrow p = 59, q = 61, \phi(n) = 3480, d = 3031$ (2020 & 2021 verbatim).
4. **Encryption of a word/letters** (2023 "CU", 2024 "DATA"): encode each letter as its ASCII code, encrypt block-by-block with $C = M^e \bmod n$ — one

square-and-multiply per letter.

5. **"Choose e"** questions (2022): pick the SMALLEST valid e with $\gcd(e, \phi(n)) = 1$ — usually $e = 5$ or $e = 7$ for small $\phi(n)$.
6. **Conceptual add-on** (2022, 8 marks): "why can't an eavesdropper decrypt without the private key" $\rightarrow$ answer = **factoring n is computationally infeasible** for large primes (RSA hardness assumption).
7. **Square-and-multiply is mandatory** for any exponent > a few — write e (or d) in binary, square-then-conditionally-multiply bit by bit. Never attempt to compute M^e directly.

This topic is built directly on Topic 7 (Extended Euclidean for d, Euler's totient for $\phi(n)$). If Topic 7 is solid, RSA is just "plug numbers into the same formula 8 times."